

Soft Prior or Hard Mask

How the nine gene-gene graphs enter the cell graph transformer: the graph-regularization sweep on the 025 triples, read at three seeds per arm

Michael Volk

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Abstract

Nine gene-gene graphs enter layer 1 of the cell graph transformer by one of two mechanisms: a soft prior, a divergence penalty of weight λ that pulls each head's attention toward its graph's normalized adjacency, or a hard mask that restricts each head's attention to its graph's edges. The sweep runs one protocol (the 376,732 triples of experiment 010 on their pinned random split, a learnable gene table, AdamW at a constant 2.5×10^{-4} , 30 epochs, four A40s per run) at $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, under the mask, and under the prior at $\lambda = 10^{-3}$ toward degree-matched random rewirings of the same graphs, three seeds per arm. Read at epoch 29, the mask sits with no penalty, 0.421 ± 0.003 against 0.425 ± 0.000 held-out Pearson, and the prior rises with λ without a peak inside the ladder, to 0.447 ± 0.001 at $\lambda = 1$, above no penalty on every seed by 0.021 to 0.023. Read at each run's best epoch, all nine arms lie within 0.010 of one another (0.444 to 0.454): the prior moves the peak by at most 0.006 and moves when it occurs, from epoch 15 or 16 without a penalty to epochs 23 to 26 at $\lambda = 1$, so most of what it buys at the fixed reading is resistance to the decline the unpenalized model shows after epoch 16. The attention contracts onto the graph at every λ tested, with edge recall at degree 0.78 at 10^{-5} and 0.915 from 10^{-2} up against 0.006 with no penalty, and saturates two decades below the λ at which accuracy is highest, so the contraction itself is not what carries the accuracy. The random-graph control at $\lambda = 10^{-3}$, two complete seeds, sits at the no-penalty level, 0.425 ± 0.011 , where the biological graphs give 0.432 ± 0.005 ; at that λ the effect is too small for two seeds to separate, and the control has not been run at the λ where the effect is largest. The hypothesis under test, that the graphs help only when the model is optimized through the divergence and not when they are imposed as a constraint, holds on the fixed-epoch reading and remains untested against the alternative that any graph of the same degree sequence would do the same.

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Contents

1	The question 	2
2	The ladder: accuracy against the prior's weight 	4
3	What the prior does to training 	6
4	The random-graph control: biology or any target 	7
5	Reading, and what is still missing 	7

1 The question x

The cell graph transformer of experiment 010 attends over all 6,607 genes in every layer. Nine gene-gene graphs, physical and regulatory interactions, TFLink, and the six STRING channels (neighborhood, fusion, co-occurrence, co-expression, experimental, database), are given to nine heads of layer 1, one graph per head. Two mechanisms can hand a graph to a head. The **soft prior** adds to the training loss, for each head k with row-normalized adjacency $\tilde{A}_k$ and attention α_k on the wild-type graph,

$$\lambda \sum_{k=1}^9 \frac{N}{|E_k|} \sum_i D_{\text{KL}}(\tilde{A}_{k,i\cdot} \parallel \alpha_{k,i\cdot}), \quad (1)$$

with N the number of genes and $|E_k|$ the edge count of graph k , so a head is pulled toward spreading each gene’s attention uniformly over that gene’s neighbors and is free to put mass anywhere. The rest of the loss is the same in every arm: the mean squared error of the z-scored interaction (the point loss, weight 1) and a Sinkhorn-regularized Wasserstein distance between the batch’s predicted and observed label distributions (weight 0.1, blur 0.05, extttgeomloss); the gradient probe shows the Wasserstein term at about a fifth of the point-loss gradient after epoch 0 and it is not varied here. The **hard mask** sets the attention logits of every non-edge (self-loops kept) to $-\infty$ before the softmax in layer 1, so a head can only attend along the graph and needs no penalty and no λ . The two are the ends of one axis: the mask is the $\lambda \rightarrow \infty$ limit of Eq. 1 on the support, but not in the weights, since the mask lets a head weight its neighbors as it likes while the prior pulls toward the uniform row of $\tilde{A}_k$.

Earlier readings put the two mechanisms far apart. The 010 model at $\lambda = 10^{-3}$ under a cosine schedule reached 0.456 to 0.464 held-out Pearson over three seeds, and on the 025 build the same configuration replicated at 0.446 (GilaHyper job 1598, W&B 0yw7moue), while the two hard-mask arms run then collapsed: 0.307 at epoch 1 and then near 0 (7f1yrsq9), or near 0 for 17 epochs (4qmgkcggn). Those mask runs were under the cosine schedule with a different normalizer, so the collapse was not a controlled comparison. The hypothesis this sweep was designed to test is stated in the form it was proposed: *the graphs help, but only when the model is optimized through the divergence; imposing them as a mask does not help*. A second question rides on the first: if the prior helps, is it the biology of the nine graphs or would any graph with the same degree sequence condition training the same way?

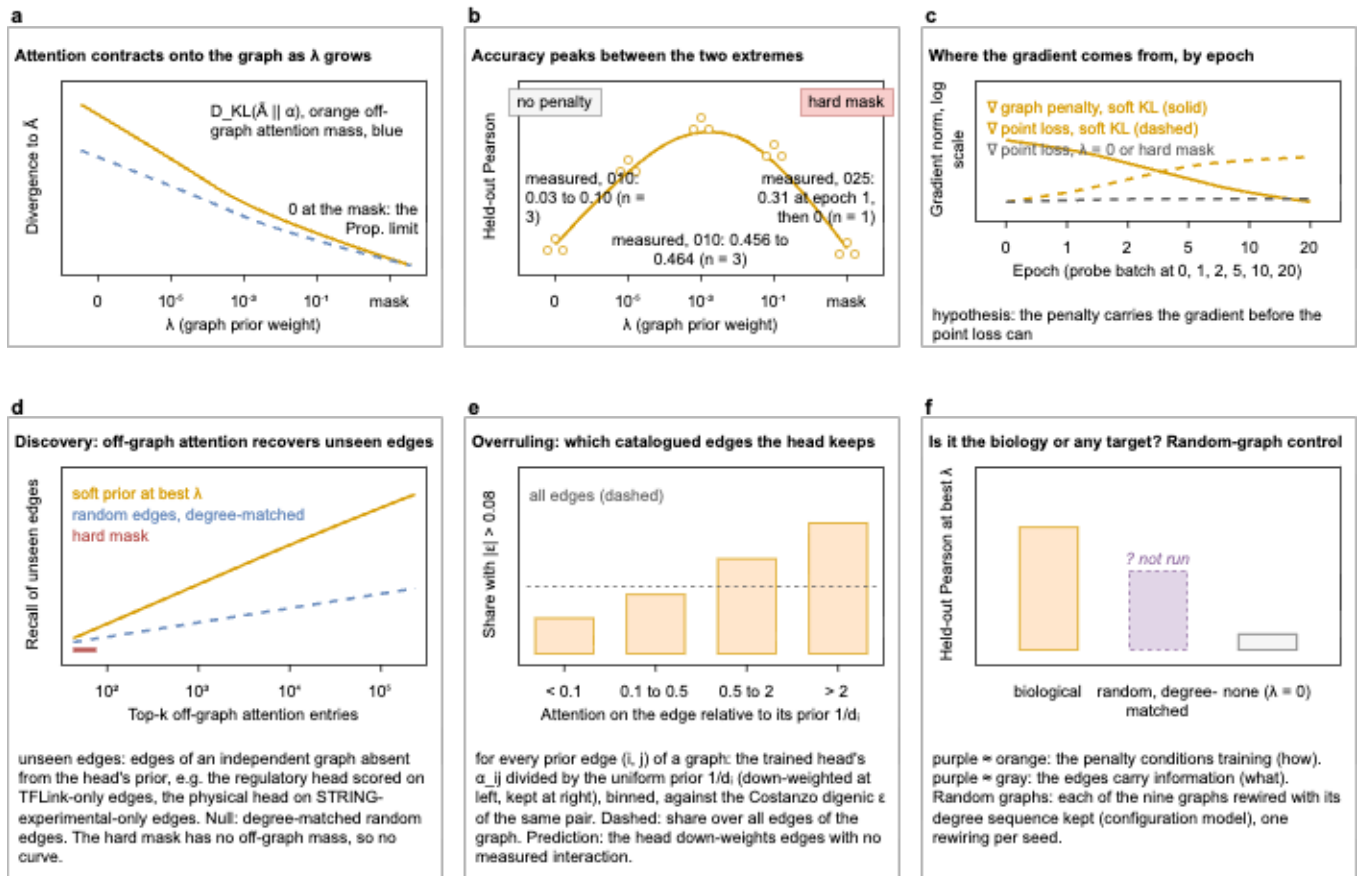
Design. Every arm is the constant-rate control `cgt_s0_r_kl_ctrl_013` with one key changed (Table 1). The control is the 376,732 triples of experiment 010 (subset S0 of the 025 build) on 010’s pinned random split, 301,386 training and 37,673 validation triples, a learnable 1,000-dimensional gene table, the label normalizer fit on the training split, AdamW at 2.5×10^{-4} with no schedule, batch 256 over four GPUs, 30 epochs. The runs are Delta `gpuA40x4` jobs, one node and four A40s per run, 24 h clocks, the 554 GB LMDB staged onto node-local NVMe at the start of each job (`experiments/025-solid-growth/scripts/delta_cgt.slurm`, `experiments/025-solid-growth/scripts/delta_submit_sweep.sh`; jobs 22055147 to 22055169, 22056267 to 22056269, 22080055, 22080057 and 22107791). The prior arms peak at 6.05 GB of allocated GPU memory per card, the mask at 2.76 GB, because only the penalized layer materializes its attention.

Table 1: The arms. Every arm is `cgt_s0_r_kl_ctrl_013` with the one change named; the control itself is the ladder’s $\lambda = 10^{-3}$ point. Complete runs are those with 30 logged epochs on W&B at the pull of 2026-09-27; a seed listed as partial was still running.

arm	change on the control	graph target	complete seeds
no penalty	<code>graph_reg_lambda=0</code>	none	1, 2, 3
prior ladder	<code>graph_reg_lambda</code> $\in \{10^{-5}, 10^{-4}, 10^{-2}, 10^{-1}, 1\}$	nine biological graphs, layer 1	1, 2, 3 each; 10^{-5} : 1, 3
control	none	nine biological graphs, layer 1, $\lambda = 10^{-3}$	1, 2, 3
hard mask	<code>cgt_s0_r_mask_028</code> : mask on, $\lambda = 0$	nine biological graphs, layer 1	1, 2, 3
random graphs	<code>cgt_s0_r_kl_rand_031</code> : rewiring on	nine rewired graphs, layer 1, $\lambda = 10^{-3}$	1, 2; 3 partial (epoch 16)

Readings. Every run is read three times. The **fixed-epoch reading** is validation Pearson at epoch 29, the protocol’s last epoch, the same optimization state for every arm. The **max-over-epochs reading** is the largest validation Pearson over the 30 epochs, an upward-biased order statistic reported with the epoch at which it occurred; it is the number a best-checkpoint deployment would see. The **minimum-loss reading** is validation Pearson at the epoch of that run’s lowest validation point loss, the epoch a loss-monitored checkpoint would keep; the loss curve is jagged after its minimum, so the epoch it picks varies by seed. Arm values are the mean and sample standard deviation over seeds; a paired comparison is by seed, since the split is pinned and the seed sets the initialization and the batch order for every arm alike.

Vocabulary. Terms used in the figures and tables, in alphabetical order.



Experiment design (Delta, gpuA40x4, 4-GPU DDP, one node per run)

Arm	λ	Graph target	Seeds	Runs	Wall clock	Panels
no penalty	0	none	3	3	12 h	b, c, f
λ sweep	$10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}$	9 biological graphs, layer 1	3 each	15	24 h	a, b, c, d, e
hard mask	∞ (mask at layer 1)	9 biological graphs	3	3	12 h	b, c, d
random-graph control	best λ from the sweep	degree-matched random	3	3	24 h	f
total				24		2,016 GPU-h

Fixed across arms: experiment-025 S0 (the 376,732 records of 010), the pinned 010 random split, 010's schedule (batch 256, bf16, OneCycle to 5e-4), one regularized layer (layer 1, all nine heads), each run read at its best-validation checkpoint. GPU-hours charge per GPU: 4 x wall clock per run. Anchors already measured: 010 $\lambda = 0$ and $\lambda = 10^{-2}$ (x367 coefficient), 025 soft KL and hard mask. Panels d and e can be computed today on the three 010 checkpoints, before any run.

Figure 1: The figure as it was designed before any run, drawn in draw.io from notes/assets/drawio/FigS-graph-regularization-sweep.gen.py before the sweep was submitted (week of 2026-09-08): sketched curves, not data. Panel a predicted contraction of attention onto the graph with λ ; b an accuracy peak between no penalty and the mask; c the penalty carrying the gradient before the point loss can; d discovery of unseen edges from off-graph attention; e overruling of catalogued edges by the digenic interaction; f the degree-matched random-graph control at the best λ . The design strip below the panels lists the arms and the Delta budget under the cosine schedule that the constant-rate protocol later replaced. Figure 2 is what was measured; Section 5 sets the two side by side.

Table 2: Controlled vocabulary.

term	meaning
divergence	Eq. 1 without its λ : the logged penalty divided by λ , summed over the nine heads and normalized by edge count, computed on the wild-type graph. Comparable across the ladder; not logged at $\lambda = 0$ or under the mask, where the model skips the computation. Reported from the validation pass at epoch 29.
edge recall at degree	For each gene i of graph k with degree d_i , the fraction of i 's neighbors among the d_i largest entries of $\alpha_{k,i}$; averaged over genes, then over the nine heads. It is 1 by construction under the mask and near $1/N$ for attention unrelated to the graph. Logged at epochs 9, 19 and 29.
gradient probe	On the first batch of epochs 0, 1, 2, 5, 10 and 20, one extra backward pass per loss term; the ℓ_2 norm of the parameter gradient of the point loss and of the penalty, before clipping, and their ratio.
point loss	Mean squared error of the prediction against the z-scored interaction label.
random graphs, degree-matched	Each of the nine graphs rewired by seeded double-edge swaps (five swaps per edge) that keep every gene's degree, in and out for the directed graphs; the rewiring seed is the run seed. The prior and the edge diagnostics are computed against the rewired graphs.
S0	The 376,732 measured triples of experiment 010, as stored in the 025 build; the pool of every arm here.

2 The ladder: accuracy against the prior's weight x

Table 3: Held-out Pearson by arm, mean over seeds ($\pm$ sample sd) at the pull of 2026-09-27, at the three readings: epoch 29, the max over epochs, and the epoch of minimum validation point loss. The highest value in each reading is bold; an arrow marks an arm whose seeds are all above ($\uparrow$) or all below ($\downarrow$) no penalty on the same reading. Generated by `graph_reg_sweep_readout.py`.

arm	n	Pearson, epoch 29	Pearson, max over epochs	Pearson, at min val loss
no penalty ($\lambda = 0$)	3	0.425 ± 0.000	0.448 ± 0.002	0.440 ± 0.005
KL $\lambda = 10^{-5}$	2	$0.420 \pm 0.002 \downarrow$	$0.444 \pm 0.002 \downarrow$	0.442 ± 0.004
KL $\lambda = 10^{-4}$	3	0.419 ± 0.010	0.446 ± 0.003	0.442 ± 0.001
KL $\lambda = 10^{-3}$ (control)	3	$0.432 \pm 0.005 \uparrow$	$0.451 \pm 0.004 \uparrow$	0.444 ± 0.003
KL $\lambda = 10^{-2}$	3	$0.438 \pm 0.002 \uparrow$	$0.453 \pm 0.002 \uparrow$	0.447 ± 0.005
KL $\lambda = 10^{-1}$	3	$0.445 \pm 0.009 \uparrow$	0.454 ± 0.006	$0.453 \pm 0.006 \uparrow$
KL $\lambda = 1$	3	$0.447 \pm 0.001 \uparrow$	$0.452 \pm 0.003 \uparrow$	0.443 ± 0.009
hard mask, layer 1	3	$0.421 \pm 0.003 \downarrow$	0.446 ± 0.003	0.442 ± 0.003
KL $\lambda = 10^{-3}$, random graphs	2 (+1 partial)	0.425 ± 0.011	0.447 ± 0.002	0.444 ± 0.008

Table 4: Diagnostics by arm, mean over complete seeds ($\pm$ sample sd): validation point loss at epoch 29 and at its minimum with the mean epoch of that minimum, training Pearson at epoch 29, the divergence of layer-1 attention from the arm's target graphs (validation pass, epoch 29), and edge recall at degree against the same target, mean over the nine heads. Generated by `graph_reg_sweep_readout.py`.

arm	point loss, ep 29	min point loss (epoch)	train Pearson, ep 29	divergence	edge recall
no penalty ($\lambda = 0$)	0.867 ± 0.012	0.801 ± 0.003 (12.0)	0.624 ± 0.006	—	0.006 ± 0.000
KL $\lambda = 10^{-5}$	0.880 ± 0.030	0.803 ± 0.003 (11.0)	0.657 ± 0.000	1540	0.782 ± 0.001
KL $\lambda = 10^{-4}$	0.878 ± 0.031	0.805 ± 0.001 (7.7)	0.663 ± 0.007	571	0.866 ± 0.001
KL $\lambda = 10^{-3}$ (control)	0.859 ± 0.007	0.798 ± 0.004 (12.0)	0.636 ± 0.004	337	0.906 ± 0.000
KL $\lambda = 10^{-2}$	0.839 ± 0.007	0.798 ± 0.004 (15.3)	0.613 ± 0.006	300	0.914 ± 0.000
KL $\lambda = 10^{-1}$	0.826 ± 0.010	0.796 ± 0.003 (17.3)	0.625 ± 0.006	297	0.915 ± 0.000
KL $\lambda = 1$	0.825 ± 0.009	0.803 ± 0.004 (18.7)	0.624 ± 0.008	298	0.915 ± 0.000
hard mask, layer 1	0.869 ± 0.015	0.799 ± 0.002 (14.0)	0.612 ± 0.001	—	—
KL $\lambda = 10^{-3}$, random graphs	0.858 ± 0.005	0.800 ± 0.004 (12.0)	0.618 ± 0.003	983	0.814 ± 0.001

The mask is no penalty. On the fixed-epoch reading the hard mask gives 0.421 ± 0.003 and no penalty 0.425 ± 0.000 (Table 3); by seed the mask is below no penalty on all three, by 0.001, 0.007 and 0.003. On the max-over-epochs reading the two are 0.446 ± 0.003 and 0.448 ± 0.002 , the mask below on two seeds of three. The mask arm also fits the training triples least, 0.612 against 0.624 train Pearson at epoch 29 (Table 4). Restricting the support of layer-1 attention to the nine graphs therefore changes nothing a validation triple can see. The collapse of the two earlier mask runs did not recur under this protocol: all three seeds train normally (Figure 2g).

The prior rises with λ and does not peak inside the ladder. At epoch 29 the ladder reads 0.420, 0.419, 0.432, 0.438, 0.445 and 0.447 at $\lambda = 10^{-5}$ through 1 (Table 3). The two smallest weights sit at or below no penalty, both seeds of 10^{-5} below all three no-penalty seeds; from 10^{-3} up every seed is above its no-penalty partner, by 0.005 to 0.013 at 10^{-3} and by 0.021, 0.023 and 0.021 at $\lambda = 1$. The mock-up drawn before the runs (`notes/assets/drawio/FigS-graph-regularization-sweep.gen.py`) placed a peak between the two extremes; the measurement places the maximum at the top of the ladder, with 10^{-1} and 1 within one standard deviation of each other (0.445 ± 0.009 , 0.447 ± 0.001). Whether the curve turns over at $\lambda = 10$ or 100 is not measured.

The max-over-epochs reading compresses the ladder. Read at each run's best epoch, the nine arms span 0.444 to 0.454, and the best arm ($\lambda = 10^{-1}$, 0.454 ± 0.006) is 0.006 above no penalty (0.448 ± 0.002). At $\lambda = 1$ the paired differences against no penalty shrink from 0.021 to 0.023 at epoch 29 to 0.007, 0.003 and 0.003 at the best epoch, still positive on all three seeds. What changes with λ is less the height of the peak than its timing: the no-penalty seeds peak at epochs 15, 16 and 16, the mask at 15, 17 and 18, $\lambda = 1$ at 23, 24 and 26 (Table 6). The fixed-epoch gain of the prior is therefore mostly the unpenalized model's decline from epoch 16 to epoch 29 (0.441 to 0.425 in the seed mean) that the prior at high λ does not show (0.435 at 15, 0.447 at 29).

The loss-selected checkpoint flattens the ladder. A loss-monitored checkpoint keeps the epoch of minimum validation point loss, which falls at epoch 12 with no penalty and drifts later with λ (15 at 10^{-2} , 17 at 10^{-1} , 19 at $\lambda = 1$; Table 4). Read there, no penalty gives 0.440 ± 0.005 , the mask 0.442 ± 0.003 , and the ladder 0.442, 0.442, 0.444, 0.447, 0.453 ± 0.006 and 0.443 ± 0.009 from 10^{-5} to 1 (Table 3). Only $\lambda = 10^{-1}$ stands above no penalty on every seed ($+0.012$, $p = 0.08$); $\lambda = 1$ does not, because its loss minimum lands late on a jagged curve (Figure 2i) at epochs where its Pearson is 0.435 to 0.452. The reading a deployment would see therefore depends on the stopping rule: a fixed budget favors the strongest prior, a loss-selected checkpoint favors 10^{-1} , and a Pearson-selected checkpoint separates nothing.

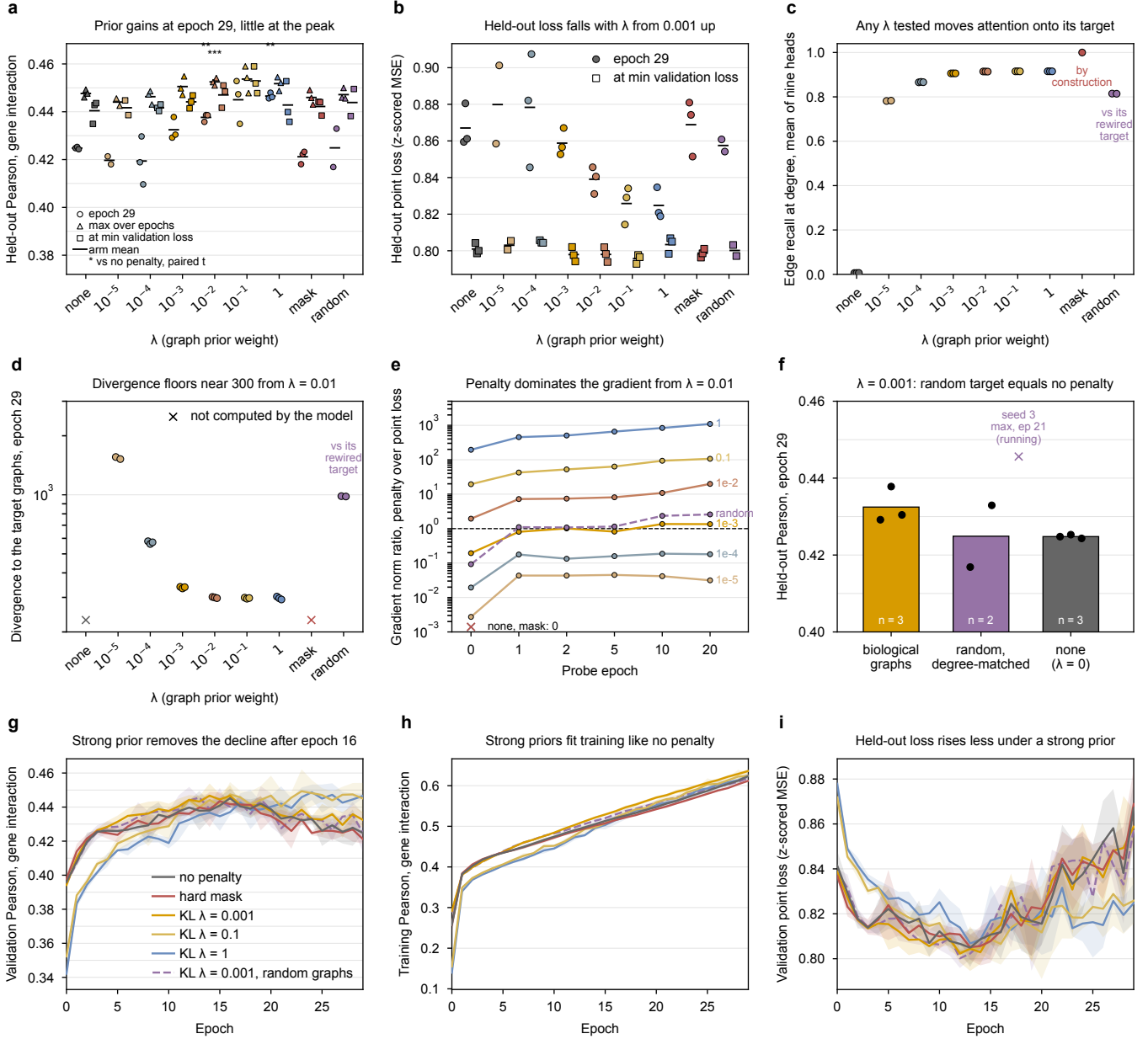


Figure 2: The sweep at three seeds per arm, read from W&B by `experiments/025-solid-growth/scripts/graph_reg_sweep_readout.py` and drawn by `experiments/025-solid-growth/scripts/graph_reg_sweep_plots.py`. One palette color per arm, kept in every panel: gray no penalty, sand $\lambda = 10^{-5}$, slate 10^{-4} , amber 10^{-3} , terracotta 10^{-2} , wheat 10^{-1} , steel blue 1, brick the hard mask, lilac the prior toward degree-matched random graphs at 10^{-3} . Every point is one seed, filled in its arm's color with a black edge; a short black bar is the arm mean; a cross marks a quantity the model does not compute. (a) Held-out Pearson on the 37,673 validation triples at three readings per seed: epoch 29 (circle, left), max over epochs (triangle, middle, every run) and at the epoch of minimum validation point loss (square, right). Stars are the paired t of each reading against no penalty on the same seeds, placed above that reading's column, one row per reading (** $p < 0.01$, *** $p < 0.001$); the minimum-loss reading has none, its closest being $\lambda = 10^{-1}$ at $p = 0.08$. (b) Held-out point loss at epoch 29 (circle) and its minimum over epochs (square), the value at the epoch the square in (a) is read at. (c) Edge recall at degree at epoch 29, mean over the nine heads, each arm against its own target: the random arm is scored against the rewired graphs it was trained toward, the mask is 1 by construction. (d) Divergence of layer-1 attention from the target graphs at epoch 29, validation pass, log scale; not computed at $\lambda = 0$ or under the mask. (e) Gradient norm of the penalty over that of the point loss on the probe batch at epochs 0, 1, 2, 5, 10 and 20, mean over seeds, log scale; the two arms without a penalty are at 0. (f) The random-graph control at $\lambda = 10^{-3}$: arm means as bars, seeds as points, the running seed's max so far as a cross. (g) Validation Pearson, (h) training Pearson and (i) validation point loss by epoch, mean over complete seeds with a $\pm$ sd band. Panel titles state the finding each panel shows and are checked against the tables in Section 5.

Paired by seed. Table 5 gives each arm minus no penalty on the same seed. With three seeds the paired t has two degrees of freedom and is reported for what it is: at epoch 29 the prior at $\lambda = 1$ is above no penalty by $+0.022 \pm 0.001$ ($t = 30$, $p = 0.001$) and at 10^{-2} by $+0.013 \pm 0.001$ ($t = 18$, $p = 0.003$); 10^{-1} is above on every seed but with a wider spread ($+0.020 \pm 0.010$, $p = 0.07$). The mask is below no penalty on every seed (-0.004 ± 0.003 , $p = 0.16$). At the best-epoch reading only 10^{-2} separates ($+0.005 \pm 0.000$, $p < 0.001$); $\lambda = 1$ is above on every seed by $+0.004 \pm 0.002$ ($p = 0.09$) and the mask is mixed. The claim the fixed-epoch reading supports at three seeds is that a strong prior beats no penalty by about 0.02; the claim it does not support is that any arm’s peak is higher than another’s by more than 0.005. The model class has never crossed 0.46 on this split under any protocol: 0.456 to 0.464 for the 010 checkpoints under the cosine schedule, 0.446 for the 025 replication (job 1598), 0.452 for the joint-fitness arm (Delta 22030924), and 0.459 for the best single run here ($\lambda = 10^{-1}$, seed 1, epoch 23).

Table 5: Each arm minus no penalty, paired by seed, on both readings. The fixed reading pairs complete runs; the max reading pairs every run (a partial run’s max is a lower bound). t is the paired t statistic on $n - 1$ degrees of freedom, p two-sided; an arrow marks a difference with the same sign on every seed. Generated by `graph_reg_sweep_readout.py`.

arm	reading	n	difference by seed	mean	paired t	p
KL $\lambda = 10^{-5}$	epoch 29	2	-0.003, -0.007	-0.0051 ↓	-2.4	0.252
KL $\lambda = 10^{-4}$	epoch 29	3	-0.005, +0.005, -0.016	-0.0054	-0.9	0.459
KL $\lambda = 10^{-3}$ (control)	epoch 29	3	+0.005, +0.013, +0.005	+0.0077 ↑	2.9	0.103
KL $\lambda = 10^{-2}$	epoch 29	3	+0.011, +0.014, +0.013	+0.0129 ↑	17.7	0.003
KL $\lambda = 10^{-1}$	epoch 29	3	+0.029, +0.023, +0.010	+0.0202 ↑	3.6	0.068
KL $\lambda = 1$	epoch 29	3	+0.021, +0.023, +0.021	+0.0218 ↑	30.0	0.001
hard mask, layer 1	epoch 29	3	-0.001, -0.007, -0.003	-0.0036 ↓	-2.2	0.160
KL $\lambda = 10^{-3}$, random graphs	epoch 29	2	-0.007, +0.008	+0.0003	0.0	0.972
KL $\lambda = 10^{-5}$	max over epochs	2	-0.002, -0.007	-0.0044 ↓	-1.9	0.313
KL $\lambda = 10^{-4}$	max over epochs	3	-0.000, +0.003, -0.006	-0.0013	-0.5	0.659
KL $\lambda = 10^{-3}$ (control)	max over epochs	3	+0.002, +0.001, +0.006	+0.0029 ↑	2.2	0.163
KL $\lambda = 10^{-2}$	max over epochs	3	+0.005, +0.005, +0.005	+0.0049 ↑	81.1	0.000
KL $\lambda = 10^{-1}$	max over epochs	3	+0.011, +0.008, -0.001	+0.0060	1.6	0.260
KL $\lambda = 1$	max over epochs	3	+0.007, +0.003, +0.003	+0.0041 ↑	3.0	0.094
hard mask, layer 1	max over epochs	3	-0.005, +0.003, -0.003	-0.0017	-0.7	0.557
KL $\lambda = 10^{-3}$, random graphs	max over epochs	3	-0.002, +0.004, -0.004	-0.0005	-0.2	0.857
KL $\lambda = 10^{-5}$	at min val loss	2	+0.001, -0.004	-0.0015	-0.6	0.670
KL $\lambda = 10^{-4}$	at min val loss	3	-0.002, +0.005, +0.000	+0.0012	0.6	0.635
KL $\lambda = 10^{-3}$ (control)	at min val loss	3	-0.002, +0.009, +0.004	+0.0037	1.1	0.380
KL $\lambda = 10^{-2}$	at min val loss	3	-0.002, +0.016, +0.006	+0.0066	1.3	0.335
KL $\lambda = 10^{-1}$	at min val loss	3	+0.015, +0.017, +0.005	+0.0124 ↑	3.3	0.080
KL $\lambda = 1$	at min val loss	3	+0.009, +0.005, -0.007	+0.0024	0.5	0.672
hard mask, layer 1	at min val loss	3	-0.005, +0.009, +0.001	+0.0018	0.4	0.714
KL $\lambda = 10^{-3}$, random graphs	at min val loss	2	-0.005, +0.015	+0.0046	0.5	0.725

3 What the prior does to training $\times$

Support contraction happens at every λ , and saturates early. Without a penalty, layer-1 attention has nothing to do with the graphs: edge recall at degree is 0.006 at every diagnostic epoch, the value of attention unrelated to the adjacency (Figure 2c). The smallest weight tested already moves it to 0.782 at epoch 29 (0.759 at epoch 9), 10^{-4} to 0.866, 10^{-3} to 0.906, and from 10^{-2} up it sits at 0.914 to 0.915 with no further change to $\lambda = 1$. The divergence tells the same story with more resolution (Figure 2d): from 1,540 at 10^{-5} through 571 and 337 to about 300 at 10^{-2} , 10^{-1} and 1 (300, 297, 298), a floor the penalty cannot push below at any weight tested. Both diagnostics are flat from 10^{-2} to 1 while the fixed-epoch accuracy still rises from 0.438 to 0.447. How close the attention sits to the graph is set by $\lambda = 10^{-2}$; what the two larger weights add is not more contraction.

The gradient budget. The gradient probe (Figure 2e) measures the norm of each loss term’s gradient on one batch before clipping. The point-loss gradient at epoch 0 is a property of the seed, not the arm: 7.4, 32.3 and 2.8 for seeds 1, 2 and 3 in every arm that shares the initialization (the mask arm, with a different layer-1 kernel, reads 5.7, 32.9 and 2.7). The penalty’s gradient scales with λ as it must, 1.1 at 10^{-3} , 10.9 at 10^{-2} , 109 at 10^{-1} and 1,097 at 1 in the seed mean at epoch 0, and grows over training while the point-loss gradient settles to 1.0 to 4.3: at epoch 20 the ratio is 1.1 to 1.5 at 10^{-3} , 15 to 24 at 10^{-2} , 86 to 146 at 10^{-1} and 680 to 1,550 at $\lambda = 1$. The two best arms of the ladder train under a total gradient that is almost entirely the penalty. The trainer clips the total gradient norm at 10, so at $\lambda \geq 10^{-1}$ every update is rescaled by two to three orders of magnitude before the optimizer sees it.

Hypothesis (untested): AdamW normalizes each parameter’s update by its running gradient scale, so a uniform rescaling of the whole gradient leaves the update nearly unchanged, and the penalty’s gradient is concentrated in the parameters that shape layer-1 attention (the layer-1 query and key projections and, through them, the gene table). Under that

reading the arms at $\lambda \geq 10^{-1}$ train the rest of the network on the point loss much as the no-penalty arm does, which is consistent with their training Pearson matching it (0.625 and 0.624 against 0.624), while layer 1 is held on the graph for the whole run. The mechanism is not measured here; a per-parameter-group norm on the probe batch would measure it.

The prior is not a fit-reducing regularizer at the weights that help. Training Pearson at epoch 29 (Figure 2h) is 0.624 with no penalty, 0.636 at 10^{-3} , 0.613 at 10^{-2} , 0.625 at 10^{-1} and 0.624 at 1, so the arms that generalize best at epoch 29 fit the training triples as well as the arm that generalizes worst. The two smallest weights fit the training set better than any other arm, 0.657 at 10^{-5} and 0.663 at 10^{-4} , and generalize no better than no penalty. The validation point loss (Figure 2i) reaches its minimum near epoch 13 in every arm (0.80 to 0.82) and then rises; at epoch 29 it is 0.867 with no penalty and 0.869 under the mask against 0.826 and 0.825 at $\lambda = 10^{-1}$ and 1. The prior at high weight does not stop the model from fitting the training triples; it slows the growth of held-out error after the point-loss minimum.

Hypothesis (untested): the late decline of the unpenalized model is layer-1 attention drifting away from any fixed structure as the gene table specializes to the training triples, and a prior strong enough to hold layer 1 in place removes that degree of freedom without touching the rest. The mask removes the same freedom in a different way, by fixing the support and leaving the weights free, and it declines exactly as no penalty does, so if this hypothesis is right the part of layer 1 that drifts is the weighting within the support, not the support.

4 The random-graph control: biology or any target x

The control arm keeps everything of the $\lambda = 10^{-3}$ control except the graphs: each of the nine is rewired before the model sees it, by double-edge swaps that keep every gene’s degree, five swaps per edge, seeded by the run seed (`cgt_s0_r_kl_rand_031`, `torchcell.graph.rewire`). The prior pulls the heads toward the rewired graphs and the edge diagnostics are scored against them. The arm was written to run at the best λ of the sweep; the sweep’s best was not known at submission, so it ran at 10^{-3} , the 010 operating point.

The heads learn the random graphs, less completely. Edge recall at degree against the rewired graphs reaches 0.814 at epoch 29 and the divergence 983 (Figure 2c, d), against 0.906 and 337 for the biological graphs at the same λ . A random graph of the same degree sequence is a target the attention can and does move toward, but it does not get as close in 30 epochs. Whether that is because the biological graphs are more compatible with the point loss, or because the rewired graphs are less clustered and so harder to represent with a low-rank attention, is not separated by anything logged.

At $\lambda = 10^{-3}$ the accuracy question is underpowered. On the fixed-epoch reading the two complete random-graph seeds give 0.417 and 0.433, against 0.429 and 0.438 for the biological graphs on the same seeds and 0.424 and 0.425 with no penalty (Figure 2f). The biological graphs are above the random ones on both seeds, by 0.012 and 0.005; the random ones straddle no penalty. At the max-over-epochs reading the three arms are 0.451, 0.447 and 0.448. The effect being tested at this λ is the 0.005 to 0.013 that the biological prior adds over no penalty at epoch 29, and two seeds of a control whose own seed spread is 0.011 cannot resolve it. The question the control was designed to answer, whether the biology or the conditioning carries the gain, is open, and it is answerable only where the gain is large: at $\lambda = 1$ the biological prior is 0.021 to 0.023 above no penalty on every seed, so three seeds of the rewired graphs at $\lambda = 1$ would separate a 0.02 effect from the 0.001 to 0.011 seed spread seen here. That arm is one override away (`model.graph_regularization.graph_reg_lambda=1` on `cgt_s0_r_kl_rand_031`) and has not been submitted.

5 Reading, and what is still missing x

On the hypothesis. Stated as proposed: the graphs help, but only when the model is optimized through the divergence; imposing them as a mask does not help. The mask half holds on both readings and all three seeds: the mask is indistinguishable from no penalty (0.421 against 0.425 at epoch 29, 0.446 against 0.448 at the best epoch), and it trains without the collapse the earlier cosine-schedule runs showed. The prior half holds on the fixed-epoch reading from $\lambda = 10^{-3}$ up, with the gain growing to 0.021 to 0.023 at $\lambda = 1$ and no turnover inside the ladder, and holds weakly on the best-epoch reading (0.003 to 0.007 at $\lambda = 1$, positive on every seed). The word “help” needs its reading attached: a best-checkpoint deployment gains at most 0.006 from the prior; a fixed-budget run gains 0.02, because the prior removes the decline after epoch 16 rather than raising the peak. What the sweep does not establish is that the help comes from the biology of the graphs. The random-graph control ran only at $\lambda = 10^{-3}$, where the effect is within two seeds’ noise, so “networks help” is measured as “a strong prior toward these graphs helps” and the alternative, that a strong prior toward any graph of the same degree sequence helps as much, is untested.

What the mock-up predicted and what was measured. The figure design (`notes/assets/drawio/``FigS-graph-regularization-sweep.gen.py`) drew six panels. Panel a, contraction with λ : measured, and it saturates at 10^{-2} . Panel b, accuracy peaking between the extremes: measured, and there is no interior peak; the maximum is at the top of the ladder. Panel c, the penalty carrying the gradient before the point loss can: measured, and the penalty dominates the gradient at every probe epoch for $\lambda \geq 10^{-2}$, not only early. Panels d and e, discovery of edges outside a head’s prior and overruling of cataloged edges by the digenic interaction, need the layer-1 attention of the saved checkpoints, which are on Delta /`scratch` and have not been read; the edge-recovery precision logged at $k = 32$ (0.21 for every prior arm from 10^{-3} up, 0.006 with no penalty) is the nearest logged quantity and is not the discovery statistic. Panel f, the random-graph control: measured at the wrong λ .

The panel titles, checked. Each title of Figure 2 states a finding, which is a place for bias to enter, so each is set against the number it rests on. (a) “Prior gains at epoch 29, little at the peak”: +0.022 at $\lambda = 1$ on the fixed reading against +0.004 on the max reading (Table 5). (b) “Held-out loss falls with λ from 0.001 up”: 0.859, 0.839, 0.826, 0.825 at 10^{-3} to 1 against 0.867 with no penalty; the two smallest weights are above no penalty (0.880, 0.878), which the title’s “from 0.001 up” carries (Table 3). (c) “Any λ tested moves attention onto its target”: 0.782 at 10^{-5} against 0.006 at 0. (d) “Divergence floors near 300 from $\lambda = 0.01$ ”: 300, 297, 298. (e) “Penalty dominates the gradient from $\lambda = 0.01$ ”: ratio 15 to 24 at 10^{-2} by epoch 20 against 1.1 to 1.5 at 10^{-3} . (f) “ $\lambda = 0.001$: random target equals no penalty”: 0.425 ± 0.011 against 0.425 ± 0.000 , two seeds, and the biological graphs at 0.432 ± 0.005 . (g) “Strong prior removes the decline after epoch 16”: seed-mean 0.441 to 0.425 with no penalty against 0.435 to 0.447 at $\lambda = 1$ and 0.441 to 0.445 at 10^{-1} . (h) “Strong priors fit training like no penalty”: 0.625 and 0.624 against 0.624. (i) “Held-out loss rises less under a strong prior”: from the epoch-13 minimum, 0.801 to 0.867 with no penalty against 0.803 to 0.825 at $\lambda = 1$.

Open items.

1. The random-graph arm at $\lambda = 1$, three seeds. The one run that decides between conditioning and biology.
2. $\lambda = 10$ and 100, three seeds each, to find the turnover the ladder did not reach. Memory is unchanged (the penalty is computed at every $\lambda > 0$).
3. Seed 2 of $\lambda = 10^{-5}$ (Delta job 22055161) has no run on W&B; the job’s log on Delta says why. The arm is read at two seeds.
4. Seed 3 of the random-graph arm (job 22056269) was at epoch 16 at the pull; the tables refresh with `make refresh`.
5. Panels d and e from the checkpoints: pull the best-Pearson checkpoint of one seed per arm from Delta and score layer-1 attention against TFLink-only and STRING-experimental-only edges and against Costanzo digenic ε .
6. A per-parameter-group gradient probe, to test the hypothesis of Section 3 about where the penalty’s gradient lands.

Table 6: Every run read, one row per seed. Epochs is the number of validation epochs logged; Pearson at epoch 29 and the max over epochs with its epoch; the gradient-probe ratio (penalty over point loss) at epoch 0; the W&B run id in project `torchcell_025-solid-growth-equivariant_cell_graph_transformer`. Generated by `graph_reg_sweep_readout.py`.

arm	seed	epochs	Pearson, ep 29	max (epoch)	gradient ratio, ep 0	W&B run
no penalty ($\lambda = 0$)	1	30	0.4243	0.4476 (16)	0.00	zvtj3waw
no penalty ($\lambda = 0$)	2	30	0.4248	0.4460 (15)	0.00	4vgeyw2i
no penalty ($\lambda = 0$)	3	30	0.4253	0.4493 (16)	0.00	mmiydys
KL $\lambda = 10^{-5}$	1	30	0.4213	0.4456 (13)	0.00	rldsrcnr
KL $\lambda = 10^{-5}$	3	30	0.4180	0.4425 (14)	0.00	7fx9njte
KL $\lambda = 10^{-4}$	1	30	0.4189	0.4473 (16)	0.01	r2i3i9xq
KL $\lambda = 10^{-4}$	2	30	0.4297	0.4486 (15)	0.00	h8zhfq2d
KL $\lambda = 10^{-4}$	3	30	0.4096	0.4431 (6)	0.04	oxh1ug5l
KL $\lambda = 10^{-3}$ (control)	1	30	0.4292	0.4498 (16)	0.14	6dvby8w
KL $\lambda = 10^{-3}$ (control)	2	30	0.4378	0.4471 (12)	0.04	ppm9kl4w
KL $\lambda = 10^{-3}$ (control)	3	30	0.4304	0.4548 (14)	0.40	22u65gax
KL $\lambda = 10^{-2}$	1	30	0.4358	0.4524 (27)	1.45	u1hdm0y0
KL $\lambda = 10^{-2}$	2	30	0.4386	0.4510 (19)	0.37	eufg8mlm
KL $\lambda = 10^{-2}$	3	30	0.4386	0.4541 (14)	4.01	0vgat9et
KL $\lambda = 10^{-1}$	1	30	0.4528	0.4590 (23)	14.47	lz1rd90p
KL $\lambda = 10^{-1}$	2	30	0.4473	0.4542 (23)	3.69	df8jt886
KL $\lambda = 10^{-1}$	3	30	0.4349	0.4478 (13)	40.20	5pz6vc5y
KL $\lambda = 1$	1	30	0.4456	0.4545 (26)	146.51	367oc8pv
KL $\lambda = 1$	2	30	0.4480	0.4487 (23)	37.17	ds928dmp
KL $\lambda = 1$	3	30	0.4461	0.4521 (24)	399.95	zxme1gip
hard mask, layer 1	1	30	0.4232	0.4430 (18)	0.00	g35yf77n
hard mask, layer 1	2	30	0.4181	0.4491 (15)	0.00	1pv21pf9
hard mask, layer 1	3	30	0.4222	0.4458 (17)	0.00	tpixauk7
KL $\lambda = 10^{-3}$, random graphs	1	30	0.4169	0.4459 (13)	0.15	usgha5w6
KL $\lambda = 10^{-3}$, random graphs	2	30	0.4329	0.4500 (19)	0.04	9tqgl66k
KL $\lambda = 10^{-3}$, random graphs	3	22 (running)	–	0.4456 (14)	0.41	zuw5z7k4